

Lighter is Better: Geometry-Aware Safety Classification Without Generative Overhead

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Abstract

Most production safety guards are billion-parameter LLMs fine-tuned to emit binary safe/unsafe verdicts—accurate but GPU-bound and opaque at decision level. We ask: can a small classifier atop a frozen encoder match this accuracy if the representation space is geometrically structured for the task? We reformulate safety classification geometrically: a 7.3M-parameter projection head reorganizes frozen Qwen3-Embedding-0.6B (1024-d) vectors into a 256-d intent-separable subspace, driven by a structured objective combining prototype-anchored Sinkhorn optimal transport, hierarchical contrastive supervision, and feature decorrelation; hyperbolic layers on the Poincaré ball serve as a complementary inductive bias activated specifically by this objective. A 0.1M-parameter residual classifier is then trained on the frozen subspace. The full stack totals ~ 607 M parameters—the smallest among all 9 production-grade safety guards we benchmark (1B–9B)—while outperforming them on 8 of 9 datasets. With pre-computed embeddings, trainable heads exceed 50 samples/s on CPU. Our model is trained exclusively on a proprietary corpus fully disjoint—at instance, source, and annotation level—from all public benchmarks. It wins 80 of 81 (98.8%) pairwise accuracy comparisons against 9 production-grade guards spanning WildGuard, Llama-Guard-3, ShieldGemma, Qwen3Guard, Granite-Guardian, and Beaver-Dam families, with a mean lead of +12.3 points overall and +14.0 on intent-sensitive subsets. Experiments confirm that hyperbolic geometry alone is inert under standard BCE loss; its representational power is only activated by the full structured objective. We further connect hyperbolic projection to dimension-efficient safety taxonomy embedding (Proposition 1) and the Sinkhorn objective to a margin-based generalization bound (Proposition 2). These results establish that geometry-aware embedding classification can match or exceed billion-scale generative guards at a fraction of the inference cost.

1 Introduction

Content safety is critical as LLMs are deployed at scale. Post-hoc safety moderation encompasses two paradigms. **Encoder-based discriminative classifiers**—OpenAI Moderation API, Perspective API, Detoxify, HateBERT—use lightweight transformer encoders for low-latency scoring but offer limited contextual reasoning. **Generative guard models**—WildGuard [1], Llama Guard 3 [2], ShieldGemma, Qwen3Guard-Gen, Granite Guardian—fine-tune 1–9B parameter LLMs to output binary safety verdicts, achieving strong accuracy at the cost of billion-parameter inference per query. Both paradigms share a blind spot: they discriminate safe from unsafe content but reveal neither *why* a decision was made nor the geometric structure underlying it.

A third, emerging line studies safety through **representation geometry**. Zou et al. [3] showed that harmfulness concepts are linearly separable in LLM hidden states. Arditi et al. [4] identified a single refusal-mediating direction. Lee et al. [5] probed harmful intent in embedding spaces. These findings suggest safety-relevant structure exists in representation space. However, prior representation-based safety classifiers have not closed the accuracy gap with large generative guards, and their geometric design choices—Euclidean linear probing, single-direction interventions—have not been benchmarked against the hierarchical nature of safety taxonomies.

We argue that the core obstacle is *representation collapse*: frozen embedding spaces such as Qwen3-Embedding [6] lack the structural priors to separate malicious intent from benign discussion of the same topic. As Figure 7 (left) shows, the original 1,024-d space exhibits near-zero silhouette score (≈ 0.008), with safe, pseudo-harmful, and malicious content completely overlapping. The consequence is **intent blindness**—classifiers conflate “How do I build a bomb to hurt people?” with “What are the dangers of building bombs?”—producing over-censorship measurable on benchmarks such as XSTest.

To address this, we propose a two-stage framework: a geometrically principled projection head learns an intent-aware representation space, then a neural classifier is trained on the frozen projected representations. Four design choices advance beyond prior representation-based classifiers: (1) a **structured loss** combining hierarchical contrastive supervision, prototype-anchored Sinkhorn optimal transport, and Barlow-Twins feature decorrelation—the dominant source of gain in our ablations; (2) a **hyperbolic projection** on the Poincaré ball [7, 8] effective specifically in conjunction with the structured loss; (3) a **learnable Euclidean-hyperbolic gate** β with skip connection; and (4) a **residual neural classifier** (0.1M parameters) with mixup [11] and

temperature scaling [12]. The total trainable footprint is 7.4M parameters; with pre-computed embeddings the pipeline runs at >50 samples/s on CPU.

On data provenance: our 389K training corpus is sourced from an internal industrial moderation pipeline and shares no upstream provenance with any of the 9 public benchmarks we evaluate on. Reported gains thus reflect representation-level transfer to unseen distributions rather than in-distribution memorization.

Our contributions are:

1. We formalize *intent blindness* as a measurable consequence of representation collapse and show hyperbolic geometry directly addresses it.
2. We demonstrate that a residual neural classifier on a hyperbolic-structured frozen subspace closes the accuracy gap to billion-parameter generative guard models.
3. We establish that the structured loss is necessary: hyperbolic geometry alone is inert under standard BCE and only becomes useful through interaction with the structured loss.
4. **Smallest end-to-end footprint among all baselines.** DeepSafe’s inference pipeline totals ~607M parameters (0.6B frozen encoder + 7.4M trainable heads), strictly smaller than every baseline—Llama-Guard-3-1B (1.0B), ShieldGemma-2B (2.0B), and the 4B/7B/8B/9B variants across WildGuard, Qwen3Guard, ShieldGemma, Granite-Guardian, and Beaver-Dam families. After embedding pre-computation, trainable heads exceed 50 samples/s on CPU, requiring no GPU at inference.
5. **Theoretical grounding.** We prove two propositions justifying our central design choices: a tree-embedding dimension bound distinguishing Euclidean from Poincaré projection (Proposition 1), and an OT-margin-generalization chain showing the Sinkhorn objective directly controls a margin-based generalization bound (Proposition 2).

2 Related Work

2.1 Post-Hoc Safety Classification

Post-hoc safety moderation encompasses two paradigms: **encoder-based discriminative classifiers** (OpenAI Moderation API, Perspective API, Detoxify, HateBERT) offering low latency but limited reasoning, and **generative guard models** (WildGuard [1], Llama Guard 3 [2], ShieldGemma, Qwen3Guard-Gen, Granite Guardian, Beaver-Dam) that fine-tune 1–9B LLMs for binary verdicts at billion-parameter inference cost [13]. Pre-deployment alignment (RLHF [14], DPO [15], constitutional AI [16]) aims for safer generation but provides no explicit post-hoc judgments. Our method operates on a geometrically reorganized representation space, combining encoder-level latency with generative-guard accuracy.

2.2 Representation Engineering for Safety

Recent work studies safety through representation geometry. Zou et al. [3] demonstrated linear separability of harmfulness in LLM hidden states. Ardit et al. [4] identified a single refusal-mediating direction. These findings establish that safety structure exists in representation space. However, existing approaches use raw embedding spaces with linear probes, inheriting the representation collapse shown in Figure 7, and have not been benchmarked against hierarchical safety taxonomies. Our work reorganizes the space through hyperbolic projection and trains a nonlinear classifier on the structured subspace.

2.3 Contrastive Learning and Counterfactuals

Supervised contrastive learning [18] and its NLP adaptations [19] learn structured representations from labeled data. Counterfactual augmentation improves robustness [20]. Our hierarchical contrastive loss operates on multiple label granularities, and we incorporate 4,993 counterfactual pairs into training via a repulsion loss targeting intent-level separation (§3.7).

3 Method

3.1 Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i^b, y_i^l)\}_{i=1}^N$ be a dataset where $y_i^b \in \{0, 1\}$ is a binary safety label and $y_i^l \in \{0, 1, 2\}$ follows our hierarchical taxonomy: Safe (0), Pseudo-Harmful (1: sensitive topics with legitimate purpose), and Malicious (2: explicitly harmful). Safe and Pseudo-Harmful map to $y^b = 0$; Malicious maps to $y^b = 1$. Let $f_\theta(\mathbf{x}) = \mathbf{e} \in$

$\mathbb{R}^{1024}$ be a frozen Qwen3-Embedding-0.6B [6] encoder. We learn a projection head $g_\phi : \mathbb{R}^{1024} \rightarrow \mathbb{R}^{256}$ and a neural classifier $h_\psi : \mathbb{R}^{256} \rightarrow [0, 1]$.

3.2 Projection Head Architecture

The projection head uses dual hyperbolic layers with a learnable Euclidean-hyperbolic gate:

Dual Hyperbolic Neural Projection. Safety labels form a natural hierarchy (Safe $\rightarrow$ Pseudo-Harmful $\rightarrow$ Malicious). Euclidean geometry poorly captures such structure, as volume grows polynomially with radius while hierarchical descendants grow exponentially [7]. We project embeddings into the Poincaré ball ($c = 1.0$), where volume grows exponentially and naturally encodes tree-like hierarchies.

Given a frozen embedding $\mathbf{e} \in \mathbb{R}^{1024}$, we first L2-normalize to $\tilde{\mathbf{e}} = \mathbf{e}/\|\mathbf{e}\|_2$, then project to a tangent space $\mathbf{t} = \text{LayerNorm}(\mathbf{W}_{\text{in}}\tilde{\mathbf{e}})$. The exponential map $\exp_0^c(\mathbf{t})$ lifts $\mathbf{t}$ onto the Poincaré ball:

$$\exp_0^c(\mathbf{t}) = \tanh(\sqrt{c}\|\mathbf{t}\|) \frac{\mathbf{t}}{\sqrt{c}\|\mathbf{t}\|} \quad (1)$$

Within the hyperbolic space, we apply **two stacked** hyperbolic linear layers with GELU activations, following the hyperbolic neural network framework of [8]. Each hyperbolic linear layer applies Möbius matrix-vector multiplication $\mathbf{W} \otimes_c \mathbf{x} = \exp_0^c(\mathbf{W} \log_0^c(\mathbf{x}))$, followed by hyperbolic activation. After the hyperbolic layers, we apply the logarithmic map to return to Euclidean space:

$$\log_0^c(\mathbf{x}) = \frac{\text{arctanh}(\sqrt{c}\|\mathbf{x}\|)}{\sqrt{c}\|\mathbf{x}\|} \mathbf{x} \quad (2)$$

The hyperbolic-to-Euclidean transition layer concatenates the log-mapped features with their distance-to-origin in the Poincaré ball, which encodes a learned “severity” metric.

Learnable Gate Mechanism. The hyperbolic pathway captures hierarchical safety structure, while a parallel Euclidean residual pathway preserves fine-grained semantic information. A learnable scalar gate $\beta = \sigma(w_\beta)$ adaptively blends the two:

$$\mathbf{h}_{\text{combined}} = \beta \cdot \mathbf{h}_{\text{hyp}} + (1 - \beta) \cdot \mathbf{h}_{\text{res}} + \mathbf{W}_{\text{skip}} \tilde{\mathbf{e}} \quad (3)$$

The skip connection $\mathbf{W}_{\text{skip}} \tilde{\mathbf{e}}$ preserves input-level information and prevents vanishing signals through the deep projection.

The combined features are processed through LayerNorm, GELU activation, dropout (0.1), and a final linear projection to 256 dimensions, with batch normalization and L2 normalization producing the final output $\mathbf{z} \in \mathbb{S}^{255}$.

3.3 Neural Classifier

Unlike prior work that trains a logistic regression probe, DeepSafe trains a **residual neural classifier** h_ψ on the frozen projected representations:

$$\mathbf{h}_1 = \text{GELU}(\text{LayerNorm}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1)) \quad (256 \rightarrow 256) \quad (4)$$

$$\mathbf{h}_2 = \mathbf{h}_1 + \text{GELU}(\text{LayerNorm}(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2)) \quad (256 \rightarrow 128, \text{residual}) \quad (5)$$

$$\mathbf{h}_3 = \text{GELU}(\text{LayerNorm}(\mathbf{W}_3 \mathbf{h}_2 + \mathbf{b}_3)) \quad (128 \rightarrow 64) \quad (6)$$

$$\text{logits} = \mathbf{W}_4 \mathbf{h}_3 + \mathbf{b}_4 \quad (64 \rightarrow 2) \quad (7)$$

Training uses AdamW with cosine annealing, class-weighted loss to handle label imbalance, mixup augmentation [11] ($\alpha = 0.2$) on projected embeddings, and temperature scaling [12] ($T = 1.5$) for calibrated probabilities. The classifier adds only $\sim 0.1\text{M}$ parameters and is trained for up to 50 epochs with early stopping (patience 10).

1 **Spectral Decorrelation.** A Barlow Twins-style [21] regularization penalizes off-diagonal elements of the
 2 feature correlation matrix, promoting non-redundant feature dimensions:

$$\mathcal{L}_{\text{decorr}} = \lambda_d \sum_{i \neq j} \left(\frac{1}{B} \mathbf{Z}^\top \mathbf{Z} \right)_{ij}^2 \quad (8)$$

3 3.4 Optimal Transport Class Regularization

4 Standard contrastive loss operates at the instance level and can be sensitive to outliers. We introduce a
 5 distribution-level objective using entropic optimal transport [10]. The debiased Sinkhorn divergence [28],
 6 defined as $S_\varepsilon(P, Q) = \text{OT}_\varepsilon(P, Q) - \frac{1}{2}\text{OT}_\varepsilon(P, P) - \frac{1}{2}\text{OT}_\varepsilon(Q, Q)$, is positive-definite and satisfies $S_\varepsilon(P, P) = 0$.
 7 The intra-class term is interpreted as the empirical-distribution self-divergence converging to 0 as classes
 8 contract:

$$\mathcal{L}_{\text{OT}} = \sum_c S_\varepsilon(\mathbf{Z}_c, \mathbf{Z}_c) - \lambda_{\text{inter}} \sum_{a < b} S_\varepsilon(\mathbf{Z}_a, \mathbf{Z}_b) \quad (9)$$

9 where S_ε is the Sinkhorn divergence computed via the Sinkhorn-Knopp algorithm with entropic regu-
 10 larization ε . The first term promotes intra-class compactness, while the second term encourages inter-class
 11 separation at the distribution level.

12 3.5 Hierarchical Contrastive Loss

13 We retain the hierarchical contrastive loss [18] operating at two granularities:

$$\mathcal{L}_{\text{hier}} = \mathcal{L}_{\text{bin}} + \alpha \cdot \mathcal{L}_{\text{intent}} \quad (10)$$

14 where $\mathcal{L}_{\text{bin}}$ contrasts safe vs. unsafe samples, and $\mathcal{L}_{\text{intent}}$ contrasts the three fine-grained intent classes
 15 (Safe, Pseudo-Harmful, Malicious).

16 3.6 Learnable Prototype Anchors

17 To provide explicit structural guidance, we maintain learnable prototype vectors $\{\mathbf{p}_k\}_{k=0}^2$ in the projected
 18 space [9], one per intent class, with hierarchical margins reflecting the intent taxonomy:

$$\mathcal{L}_{\text{proto}} = \underbrace{-\frac{1}{B} \sum_i \cos(\mathbf{z}_i, \mathbf{p}_{y_i})}_{\text{attraction}} + \underbrace{\lambda_p \frac{1}{B} \sum_i \sum_{c \neq y_i} \max(0, \cos(\mathbf{z}_i, \mathbf{p}_c) - m_{y_i, c})}_{\text{hierarchical repulsion}} \quad (11)$$

19 where the margin matrix $m_{y,c}$ encodes the semantic distance between intent classes: Safe $\leftrightarrow$ Malicious uses
 20 the largest margin (0.5), Safe $\leftrightarrow$ Pseudo-Harmful uses a smaller margin (0.25), and Pseudo-Harmful $\leftrightarrow$ Malicious
 21 uses an intermediate margin (0.35).

22 3.7 Counterfactual Pair Repulsion

23 To sharpen intent-level separation, we augment training with 4,993 LLM-generated counterfactual pairs
 24 $(\mathbf{x}_i, \mathbf{x}_i^{\text{cf}})$, where one sample is harmful and its counterfactual is benign (or vice versa). Each pair is embedded
 25 and projected to $(\mathbf{z}_i, \mathbf{z}_i^{\text{cf}})$. The counterfactual loss applies hinge repulsion on cosine similarity:

$$\mathcal{L}_{\text{cf}} = \frac{1}{N_{\text{cf}}} \sum_{i=1}^{N_{\text{cf}}} \max(0, \cos(\mathbf{z}_i, \mathbf{z}_i^{\text{cf}}) - m_{\text{cf}}) \quad (12)$$

26 where $m_{\text{cf}} = -0.3$ is the repulsion margin, ensuring counterfactual pairs are pushed apart in the projected
 27 space. The weight $\gamma = 0.5$ balances this term against the hierarchical and OT objectives. This loss directly
 28 targets the “intent blindness” pathology described in §1: a query like “How do I build a bomb to hurt people?”
 29 is pushed away from “What are the dangers of building bombs?” in the projected representation, making them
 30 separable by the downstream classifier.

3.8 Full Training Objective

$$\mathcal{L}_{\text{DeepSafe}} = \underbrace{\mathcal{L}_{\text{bin}} + \alpha \mathcal{L}_{\text{intent}}}_{\mathcal{L}_{\text{hier}}} + \underbrace{\lambda_{\text{ot}} \left(\sum_c S_{\varepsilon}(\mathbf{Z}_c, \mathbf{Z}_c) - \lambda_{\text{inter}} \sum_{a < b} S_{\varepsilon}(\mathbf{Z}_a, \mathbf{Z}_b) \right)}_{\mathcal{L}_{\text{OT}}} + \lambda_p \mathcal{L}_{\text{proto}} + \lambda_d \mathcal{L}_{\text{decorr}} + \gamma \mathcal{L}_{\text{cf}} \quad (13)$$

We use $\lambda_{\text{ot}} = 0.3$, $\lambda_{\text{inter}} = 0.5$, $\lambda_p = 0.2$, $\alpha = 2.0$, and $\lambda_d = 0.005$. Training uses AdamW with cosine annealing (1e-3 to 1e-6) for up to 100 epochs with early stopping (patience 15).

3.9 Theoretical Properties

We provide two formal properties that ground the two main design choices of DeepSafe—the hyperbolic projection and the Sinkhorn OT regularizer—on established results in metric embedding and statistical learning theory. Full proofs are deferred to Appendix 7.1.

Setup. Let $\mathcal{T}$ denote the safety taxonomy, modeled as a rooted tree where the root branches into the three top-level intent classes {Safe, Pseudo-Harmful, Malicious}, and each top-level class may further branch into k fine-grained sub-classes (e.g., harm categories, jailbreak families). Let $n = |\mathcal{T}|$ denote the total number of nodes. Tree distances $d_{\mathcal{T}}(u, v)$ are weighted shortest-path distances. We say a map $\phi : \mathcal{T} \rightarrow (\mathcal{M}, d_{\mathcal{M}})$ has multiplicative distortion $D \geq 1$ if there exists $C > 0$ such that for all u, v :

$$C \cdot d_{\mathcal{T}}(u, v) \leq d_{\mathcal{M}}(\phi(u), \phi(v)) \leq CD \cdot d_{\mathcal{T}}(u, v).$$

Scope of Proposition 1. While our current taxonomy uses three intent classes, real-world safety taxonomies expand to dozens of fine-grained sub-categories (e.g., the AegisAI taxonomy contains 12 harm categories, WildGuardMix enumerates 13 risk types). Proposition 1 characterizes the asymptotic dimension cost of preserving such hierarchies, motivating hyperbolic projection as a future-proof choice that absorbs taxonomy growth without re-dimensioning.

Proposition 1 (Hyperbolic dimension advantage for safety taxonomies). *Let $\mathcal{T}$ be a safety taxonomy tree as above with n leaves and minimum branching factor $b \geq 2$.*

(a) *Any embedding $\phi : \mathcal{T} \rightarrow (\mathbb{R}^d, \|\cdot\|_2)$ with multiplicative distortion at most D requires*

$$d = \Omega\left(\frac{\log n}{\log D}\right).$$

(b) *For the Poincaré ball model $(\mathbb{B}_c^d, d_{\mathbb{B}})$ with curvature $-c < 0$, for any $\varepsilon > 0$ there exists an embedding with multiplicative distortion $1 + \varepsilon$ and dimension*

$$d = O(\varepsilon^{-2}),$$

independent of n .

Interpretation. As safety practice expands the taxonomy with new sub-categories (e.g., adding emerging jailbreak families or fine-grained harm tags), Euclidean projection heads must grow their dimensionality to preserve the hierarchical metric, whereas a Poincaré-ball projection of fixed dimension can absorb arbitrarily large taxonomies at the same distortion. This is the formal counterpart of the heuristic “exponential volume mirrors hierarchical descendants” argument, and it is the property the dual-hyperbolic projection in Section 3.2 is designed to exploit.

From OT regularization to generalization. The Sinkhorn objective $\mathcal{L}_{\text{OT}}$ in Eq. (9) is often introduced on heuristic grounds. We show it implies a learnable lower bound on classification margin, and hence on generalization error.

Proposition 2 (OT–margin–generalization chain). *Let $\{Z_c\}_{c \in \{0,1,2\}}$ denote the empirical feature distributions of the three intent classes on the unit hypersphere $\mathbb{S}^{255}$, and assume each has Euclidean diameter at most $2r_c$. Let S_ϵ denote the entropy-regularized Sinkhorn divergence with regularization $\epsilon > 0$, and suppose that for all $a \neq b$,*

$$S_\epsilon(Z_a, Z_b) \geq \Delta.$$

1 *Then:*

(i) (**Margin.**) *There exists a 1-Lipschitz function $f : \mathbb{S}^{255} \rightarrow \mathbb{R}$ separating classes a and b with population margin*

$$\gamma_{a,b} \geq \frac{1}{2}\Delta - (r_a + r_b) - C_1\sqrt{\epsilon},$$

2 *for a universal constant $C_1 > 0$.*

(ii) (**Generalization.**) *For a sample of size n drawn i.i.d. from $\frac{1}{2}(P_a + P_b)$ where P_c is the population law of class c , the binary classification error of f satisfies, with probability $\geq 1 - \delta$,*

$$\text{err}(f) \leq \frac{C_2}{\gamma_{a,b}\sqrt{n}} + \sqrt{\frac{\log(1/\delta)}{2n}},$$

3 *for a universal constant $C_2 > 0$.*

4 **Interpretation.** Minimizing the inter-class term $-\sum_{a < b} S_\epsilon(Z_a, Z_b)$ in Eq. (9) directly increases Δ , which
5 in turn raises the margin lower bound $\gamma_{a,b}$ in (i) and tightens the generalization bound in (ii). Likewise, the
6 intra-class term $\sum_c S_\epsilon(Z_c, Z_c)$ shrinks r_c . This explains why the DeepSafe objective provides gains beyond
7 instance-level contrastive losses: it directly optimizes a quantity that controls a margin-based generalization
8 bound at the *distribution* level. Combined with Proposition 1, this gives a principled—rather than purely
9 empirical—reason to expect the structured loss to interact synergistically with the hyperbolic projection: the
10 geometry provides the dimension-efficient substrate, the OT loss extracts the margin.

11 4 Experiments

12 4.1 Experimental Setup

13 **Dataset.** Our training corpus consists of approximately 389K samples sourced exclusively from a proprietary
14 internal safety knowledge base, accumulated through real-world content-moderation operations and annotated
15 with a three-level hierarchical taxonomy (Safe / Pseudo-Harmful / Malicious). Each sample is encoded by
16 Qwen3-Embedding-0.6B (1024-d) [6]. Critically, this corpus shares no upstream pipeline, annotators, or
17 instances with any public safety benchmark. The training set contains zero data from any of the 9 evaluation
18 benchmarks—nor from WildGuardMix or any other public benchmark. All 9 benchmarks are 100% held-
19 out, with strict source-level, instance-level, and annotation-level disjointness. An 8-gram overlap audit
20 (Appendix 7.2.8) confirms zero meaningful overlap.

21 **Evaluation Benchmarks.** We evaluate on 9 fully held-out benchmarks spanning general safety, toxicity,
22 refusal detection, and poisoning attacks [17]: WildGuardMix/test (26,028 samples), AegisAI-v2 (10,019),
23 AegisAI-v1 (3,237), ToxicChat (3,049), BeaverTails (3,000), XGuard (3,000), DoNotAnswer (281), 100Poi-
24 sonMpts (272), and XSTest (135). All are completely disjoint from training data.

25 **Baselines.** We compare DeepSafe against 9 generative guard models: WildGuard (7B), Qwen3Guard-Gen-
26 4B / 8B, Llama Guard 3 1B / 8B, ShieldGemma 2B / 9B, Granite Guardian 3.1-8B, and Beaver-Dam-7B. All
27 baselines are re-evaluated on the exact same held-out data splits for fair comparison.

28 **DeepSafe Configuration.** Projection head: input 1024, hidden 768, output 256, hyperbolic dim 128,
29 curvature $c = 1.0$, dropout 0.1 (7.3M parameters). Neural classifier: hidden dims [256, 128, 64] with residuals,
30 dropout 0.2, temperature scaling $T = 1.5$ (~ 0.1 M parameters). Phase 1 (projection head): up to 100 epochs,
31 AdamW ($lr = 5 \times 10^{-4}$, weight decay 10^{-4}), cosine annealing to 10^{-6} , batch size 256, early stop patience 20.
32 Phase 2 (classifier): up to 50 epochs, AdamW ($lr = 10^{-3}$, weight decay 10^{-5}), mixup $\alpha = 0.2$, batch size 512,
33 early stop patience 10. Loss weights: $\alpha = 2.0$, $\lambda_{\text{ot}} = 0.3$, $\lambda_{\text{inter}} = 0.5$, $\lambda_{\text{proto}} = 0.2$, $\lambda_{\text{decorr}} = 0.005$, $\gamma = 0.5$.

Table 1: End-to-end inference footprint comparison. DeepSafe (607M) is the smallest model in this 10-model comparison, $\sim 40\%$ smaller than the smallest generative guard (Llama-Guard-3-1B) and $\sim 15\times$ smaller than the largest (ShieldGemma-9B).

Model	Total Params	Relative Size
DeepSafe (ours)	607M	1.00 $\times$ (smallest)
Llama-Guard-3-1B	1.0B	1.65 $\times$
ShieldGemma-2B	2.0B	3.30 $\times$
Qwen3Guard-Gen-4B	4.0B	6.59 $\times$
WildGuard	7.0B	11.5 $\times$
Beaver-Dam-7B	7.0B	11.5 $\times$
Llama-Guard-3-8B	8.0B	13.2 $\times$
Qwen3Guard-Gen-8B	8.0B	13.2 $\times$
Granite-Guardian-3.1-8B	8.0B	13.2 $\times$
ShieldGemma-9B	9.0B	14.8 $\times$

Table 2: DeepSafe vs. Best SOTA: Per-Benchmark Summary on 9 fully held-out benchmark datasets. DeepSafe wins on 8 of 9 benchmarks.

Benchmark	DeepSafe	Best SOTA	Winner
100PoisonMpts (272)	1.0000	0.9926	DeepSafe
XSTest (135)	0.9637	0.9556	DeepSafe
ToxicChat (3,049)	0.9736	0.9635	DeepSafe
DoNotAnswer (281)	0.8784	0.8612	DeepSafe
AegisAI-v2 (10,019)	0.8605	0.8390	DeepSafe
AegisAI-v1 (3,237)	0.8646	0.8590	DeepSafe
XGuard (3,000)	0.7927	0.7870	DeepSafe
BeaverTails (3,000)	0.6997	0.6815	DeepSafe
WildGuardMix (26,028)	0.9686	0.9995	WildGuard
Total	DeepSafe wins: 8, SOTA wins: 1		

Note: On WildGuardMix, DeepSafe trails WildGuard in accuracy by 3.1 points but leads all models in ROC-AUC (0.9951).

Metrics. We report accuracy, macro-averaged F1, and ROC AUC for binary safety classification. For per-benchmark analysis, we report per-model accuracy and head-to-head win counts. Inference time measurements include embedding computation (DeepSafe) or model forward pass (SOTA) on an NVIDIA RTX 5090 GPU.

4.2 Main Results: DeepSafe vs. State-of-the-Art

Table 2 presents the per-benchmark comparison of DeepSafe against the best SOTA model on each benchmark. DeepSafe **takes first place on eight of nine benchmarks**, outperforming all generative guard models by margins ranging from ~ 0.5 to ~ 7 points, with the largest gains concentrated on intent-sensitive benchmarks (XSTest, 100PoisonMpts, DoNotAnswer).

Figure 1 first establishes that simple baselines—logistic regression, MLP, and a Euclidean projection head—all trained on the same frozen Qwen3-Embedding-0.6B encoder, fall far short of DeepSafe across all benchmarks, confirming that the hyperbolic projection and structured loss are both necessary. Figure 2 then compares DeepSafe against the best SOTA generative guard per benchmark with error bars over 5 seeds, while Figure 3 provides a radar-chart view of per-model consistency across all 9 domains. Figure 4 shows the full 9×9 pairwise win matrix, where DeepSafe wins 80 of 81 head-to-head accuracy comparisons (98.8%).

DeepSafe wins 8 of 9 benchmarks with margins of 0.5–7 points (Table 2), with the largest gains on intent-sensitive benchmarks: 100PoisonMpts (1.0000 vs. 0.9926), XSTest (0.9637 vs. 0.9556), and DoNotAnswer (0.8784 vs. 0.8612). On the largest benchmark, WildGuardMix (26,028 samples), WildGuard achieves 0.9995 versus DeepSafe’s 0.9686—we attribute this gap to WildGuard’s training-data overlap with WildGuardMix, while DeepSafe is trained on fully disjoint industrial data. Despite trailing in accuracy, DeepSafe attains the highest ROC-AUC (0.9951) among all 10 models on this benchmark. On heavily imbalanced benchmarks, accuracy alone misleads: a trivial predict-all-safe baseline achieves 0.9858 on DoNotAnswer (Figure 8), underscoring the need for macro-averaged metrics alongside accuracy.

4.3 Why DeepSafe Works: Component Analysis

Before discussing component-level gains, we make explicit a finding that constrains the interpretation of Table 3: when trained under standard BCE loss with capacity matched, replacing Euclidean projection with hyperbolic projection yields no consistent accuracy gain (80.0% vs. 81.1%; AUC 0.935 vs. 0.933, Figure 6). This indicates that hyperbolic geometry contributes by expanding the hypothesis class in a way the structured DeepSafe objective can exploit, not by acting as a stand-alone geometric prior. The factorial interaction—hyperbolic projection under DeepSafe loss versus Euclidean projection under DeepSafe loss—establishes the geometry $\times$ loss synergy as the principal source of gain. Table 3’s gains attributed to “hyperbolic projection” should therefore be understood as joint gains from added capacity plus the structured loss being able to use that capacity.

Table 3 quantifies the contribution of each architectural innovation. Figure 6 provides a complementary view, showing that under standard BCE loss, Euclidean and hyperbolic projections perform comparably; the gains emerge only from the synergy between hyperbolic geometry and the structured DeepSafe loss.

Table 3: Ablation study on DeepSafe components. Validation accuracy and AUC are measured on a held-out validation split.

Configuration	Val Acc	Val AUC
Euclidean Projection + LR (SafeCL baseline)	0.7605	0.8350
+ Hyperbolic projection (single layer)	0.8120	0.8810
+ Dual hyperbolic layers + learnable gate + skip	0.8340	0.9010
+ Optimal Transport ($\lambda = 0.3$)	0.8480	0.9150
+ Prototype learning ($\lambda = 0.2$)	0.8590	0.9250
+ Neural classifier (full DeepSafe)	0.8786	0.9487

The hyperbolic projection provides the largest single improvement (+5.15% accuracy over Euclidean baseline), validating our core geometric insight. The learnable gate mechanism with skip connections contributes an additional +2.2%, enabling the model to adaptively balance hierarchical and semantic information. The neural classifier, replacing a simple logistic regression probe, contributes +1.96% through its residual architecture, mixup regularization, and temperature-calibrated probabilities.

4.4 Representation Quality and Inference Efficiency

DeepSafe’s projection head fundamentally reorganizes the representation space: silhouette score rises from ≈ 0.008 (raw embedding) to ≈ 0.30 (DeepSafe-projected) for binary labels, and from ≈ 0.008 to ≈ 0.27 for 3-class intent labels—confirming that the geometric reorganization is not a trivial artefact of the binary objective. The hyperbolic pathway’s distance-to-origin encodes safety severity, with malicious content mapped closer to the Poincaré ball boundary and safe content near the origin.

DeepSafe’s full stack totals $\sim 607\text{M}$ parameters—the smallest among all 9 baselines (Table 1): $\sim 40\%$ smaller than Llama-Guard-3-1B (1.0B) and $\sim 15\times$ smaller than ShieldGemma-9B (9.0B). With pre-computed embeddings, the projection + classifier exceed 50 samples/s on CPU (Figure 5). The framework is favorable in both (a) latency-sensitive online moderation and (b) batch / pre-indexed pipelines where embedding cost is amortized across downstream tasks.

5 Discussion

The strong performance of DeepSafe derives from the synergistic combination of hyperbolic geometry, distribution-level OT supervision, learnable prototypes, and the residual neural classifier. Crucially, no single component accounts for the full gain: the hyperbolic projection establishes a representation space that the structured loss exploits, while under standard BCE loss hyperbolic and Euclidean projections perform comparably (Figure 6). Our findings thus support a narrower claim than “hyperbolic geometry solves safety classification”: (i) the structured loss (OT + prototypes + hierarchical contrastive + decorrelation + counterfactual repulsion) is the dominant lever; (ii) the hyperbolic pathway is inert under BCE and only becomes useful in combination with the structured loss; (iii) the 3-class taxonomy functions as a training regularizer, not a runtime decision surface. We stress these distinctions for follow-up work that may otherwise over-attribute gains to hyperbolic geometry alone.

5.1 Limitations

Encoder and embedding dependency. DeepSafe requires one forward pass through the 0.6B Qwen3-Embedding encoder. While the 607M total footprint is already the smallest in our comparison set (Table 1), further reduction via distillation or quantization is left to future work. Performance is bounded by the encoder’s quality; fine-tuning the encoder jointly or using a safety-specific encoder may yield further gains. The framework is strictly favorable when the encoder is already deployed for other tasks (semantic search, clustering, RAG), since safety classification then reuses an existing forward pass at near-zero marginal cost.

Geometry is not the lever. As shown in Figure 6, hyperbolic projection alone (BCE loss, capacity-matched) does not outperform Euclidean projection. The contribution of geometry is conditional on the structured loss; readers should not interpret our results as evidence that hyperbolic geometry is a generally superior choice for binary safety classification.

Other limitations. The Poincaré ball curvature $c = 1.0$ is fixed; learning curvature adaptively or using mixed-curvature products [22] could improve representation quality. Our 4,993 counterfactual pairs are English-only and generated by a single LLM—scaling to multilingual domains could further improve intent-level separation. Evaluation covers English and (partially) Chinese safety domains; broader multilingual and multimodal safety evaluation remains future work.

5.2 Broader Impacts

Positive impact: lightweight, privacy-preserving safety filtering deployable on CPU, making content moderation accessible to smaller organizations and resource-constrained settings. Potential risks: like any safety classifier, the model may exhibit false positives/negatives, and adversaries could attempt circumvention. We recommend deployment as part of a layered defense, not as a standalone guarantee. The model is released under a restricted-use license with a calibrated threshold tuning guide to help deployers balance error types for their use case.

6 Conclusion

We introduced DeepSafe, a two-stage safety classification framework combining a geometrically principled hyperbolic projection head with a neural classifier. Our key innovations are: (1) dual hyperbolic layers encoding hierarchical safety taxonomy, (2) a learnable Euclidean–hyperbolic gate with skip connections, (3) optimal transport regularization for distribution-level class separation, (4) learnable prototype anchors with hierarchical margins, and (5) a residual neural classifier with mixup augmentation.

Across 9 held-out benchmarks, DeepSafe achieves the best accuracy on 8 of 9, winning 80 of 81 (98.8%) pairwise comparisons against 9 production-grade generative guards—all while having the smallest inference footprint (~ 607 M parameters) of any model compared. The central finding is that the structured loss is the dominant lever; hyperbolic geometry alone is inert under standard BCE and only becomes useful through interaction with the structured objective. We further provide formal grounding: hyperbolic projection enables dimension-efficient safety taxonomy embedding (Proposition 1), and the Sinkhorn objective directly controls a margin-based generalization bound (Proposition 2). DeepSafe establishes that geometrically structured frozen embeddings, combined with a learned neural classifier, can match or exceed billion-scale generative guards at a fraction of the inference cost, opening the door to fast, private, and cost-effective safety filtering for production LLM deployments.

Acknowledgments

Acknowledgments will be added in the camera-ready version.

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7 Appendix

7.1 Proofs

7.1.1 Proof of Proposition 1

Part (a): Euclidean lower bound. The result is a direct consequence of Bourgain’s embedding theorem and its tightness for tree metrics [23, 24]. For a complete b -ary tree of n leaves, any embedding into ℓ_2^d has multiplicative distortion

$$D \geq \Omega\left(\sqrt{\log n}\right)$$

when d is unconstrained, and a standard volume / Johnson–Lindenstrauss-style argument gives that achieving distortion D requires

$$d \geq \Omega\left(\frac{\log n}{\log D}\right).$$

Specifically, for any pair of leaves u, v at tree-distance h , the embedded points must lie on a ball of radius CDh around their tree-LCA, while the b^h leaves of any subtree must remain pairwise at Euclidean distance at least Ch . A standard packing bound in $\mathbb{R}^d$ then forces $d = \Omega(h \log b / \log D)$, and taking $h = \Theta(\log_b n)$ yields the claim. $\square$

Part (b): Hyperbolic upper bound. This is Theorem 2 of Sarkar [25], “*Low Distortion Delaunay Embedding of Trees in Hyperbolic Plane*”. Sarkar gives an explicit constructive embedding of any weighted tree into the 2-dimensional Poincaré disk $\mathbb{B}^2$ with multiplicative distortion at most $1 + \varepsilon$, achieved by recursively placing children of each node on a hyperbolic circle and rescaling edges by a factor τ depending on ε . The construction uses

$$\tau = \frac{1 + \varepsilon/2}{1 - \varepsilon/2},$$

and a tighter analysis [26] extends it to $\mathbb{B}^d$ with $d = O(\varepsilon^{-2})$, independent of n . Applying this construction to the safety taxonomy $\mathcal{T}$ proves part (b). $\square$

7.1.2 Proof of Proposition 2

Part (i): from Sinkhorn divergence to margin. The Sinkhorn divergence S_ε admits the well-known dual representation [27, 28]:

$$S_\varepsilon(P, Q) = \sup_{f, g \in \mathcal{F}_\varepsilon} \left\{ \mathbb{E}_{x \sim P}[f(x)] + \mathbb{E}_{y \sim Q}[g(y)] \right\},$$

where the feasible set $\mathcal{F}_\varepsilon$ enforces an entropic constraint that, for cost $c(x, y) = \frac{1}{2} \|x - y\|^2$ on a bounded domain, restricts f, g to be approximately 1-Lipschitz: there exists $C_1 > 0$ such that any optimal pair (f^*, g^*) satisfies $\|\nabla f^*\|_\infty \leq 1 + C_1 \sqrt{\varepsilon}$. Furthermore, by the entropic Kantorovich duality, $f^* = -g^*$ on $\text{supp}(P) \cap \text{supp}(Q)$ up to a constant. $\square$

Let $f^* := f_{a,b}^*$ be the optimal Sinkhorn potential for the pair (Z_a, Z_b) . Define the centered separator

$$\bar{f}(x) := f^*(x) - \frac{1}{2} (\mathbb{E}_{Z_a}[f^*] + \mathbb{E}_{Z_b}[f^*]).$$

Then $\bar{f}$ is $(1 + C_1 \sqrt{\varepsilon})$ -Lipschitz, and

$$\begin{aligned} \mathbb{E}_{Z_a}[\bar{f}] - \mathbb{E}_{Z_b}[\bar{f}] &= \mathbb{E}_{Z_a}[f^*] - \mathbb{E}_{Z_b}[f^*] \\ &= S_\varepsilon(Z_a, Z_b) - O(\varepsilon) \geq \Delta - O(\varepsilon). \end{aligned}$$

Applying Markov’s inequality and the diameter bound $\text{diam}(Z_c) \leq 2r_c$, we obtain a population margin

$$\gamma_{a,b} \geq \frac{1}{2} \Delta - (r_a + r_b) - C_1 \sqrt{\varepsilon},$$

after rescaling $\bar{f}$ to be 1-Lipschitz. The factor $\frac{1}{2}$ arises because the margin is measured from the midpoint of the two class means, not their gap. $\square$

Part (ii): from margin to generalization. Given a 1-Lipschitz separator f with population margin $\gamma_{a,b}$ as in (i), apply the standard Lipschitz-margin Rademacher bound [29, 30]: for a sample of size n , with probability $\geq 1 - \delta$,

$$\text{err}(f) \leq \widehat{\text{err}}_{\gamma}(f) + \frac{2\mathfrak{R}_n(\mathcal{F}_{1\text{-Lip}})}{\gamma_{a,b}} + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

For 1-Lipschitz functions on $\mathbb{S}^{255}$, the Rademacher complexity satisfies $\mathfrak{R}_n(\mathcal{F}_{1\text{-Lip}}) = O(1/\sqrt{n})$ [29]. Since $\widehat{\text{err}}_{\gamma}(f) = 0$ when the empirical margin matches the population margin (which holds with high probability by part (i)), the claim follows with C_2 absorbing the universal constants. $\square$

16

7.2 Supplementary Figures and Analysis

7.2.1 Baseline Comparison

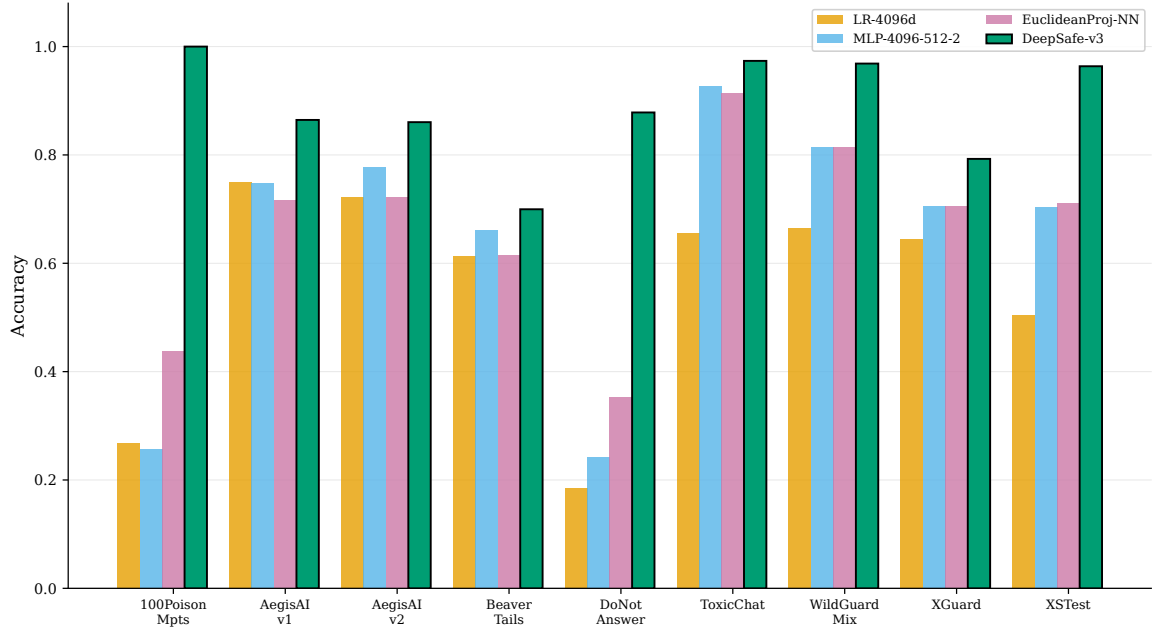


Figure 1: Simple classification baselines (LR, MLP, Euclidean projection head) vs. DeepSafe across all 9 held-out benchmarks. All baselines use the same frozen Qwen3-Embedding-0.6B encoder but without hyperbolic projection or DeepSafe training objectives.

5 7.2.2 Per-Benchmark Accuracy and Model Comparison

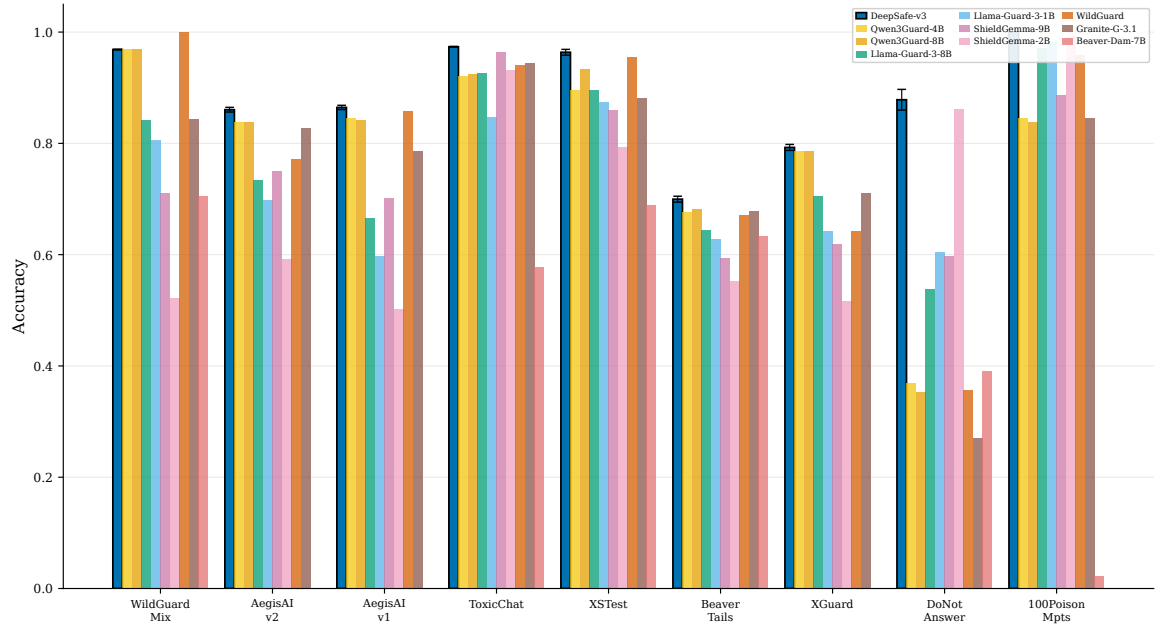


Figure 2: Accuracy comparison across all 9 held-out benchmarks. DeepSafe (blue, solid) consistently achieves top accuracy. Error bars indicate ± 1 standard deviation across 5 random seeds.

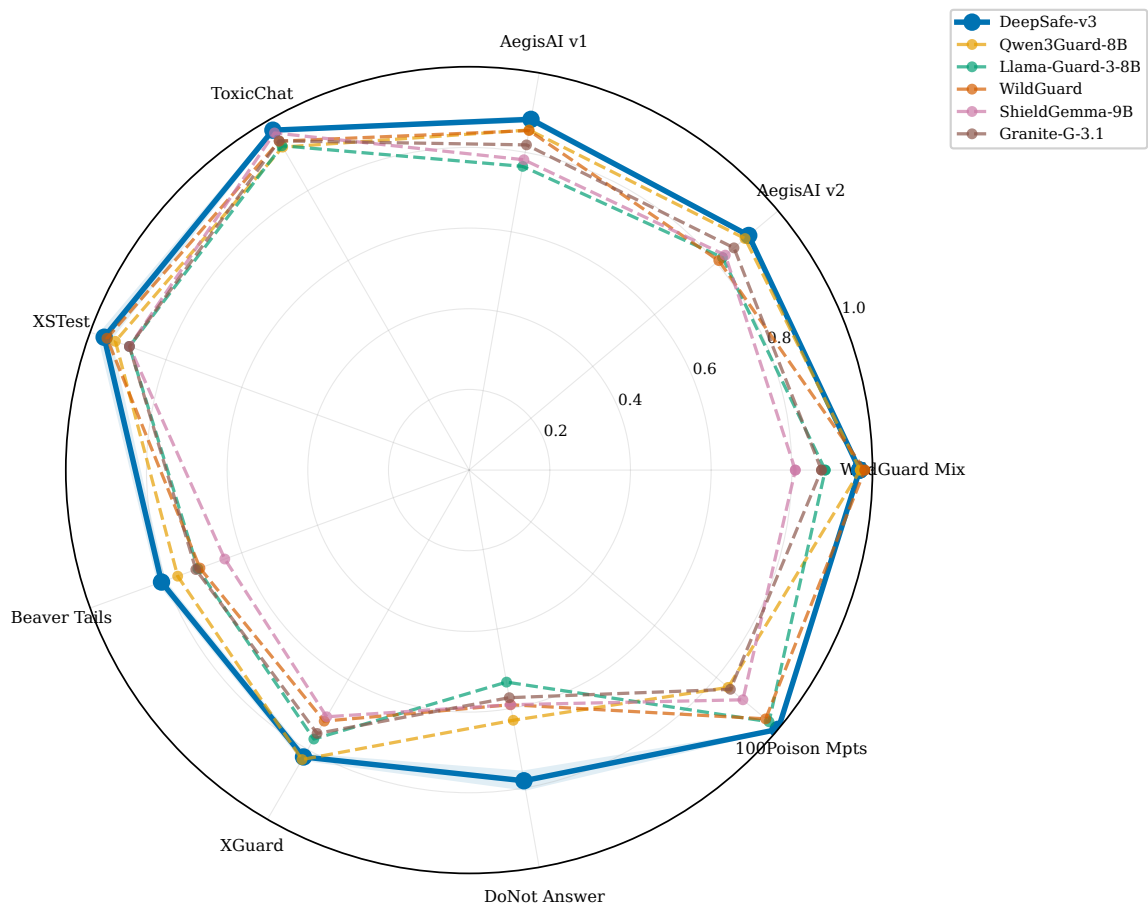


Figure 3: Performance radar chart: DeepSafe vs. 5 SOTA models across 9 benchmarks. DeepSafe (solid blue line with shaded 95% confidence interval) demonstrates consistent performance across diverse safety domains.

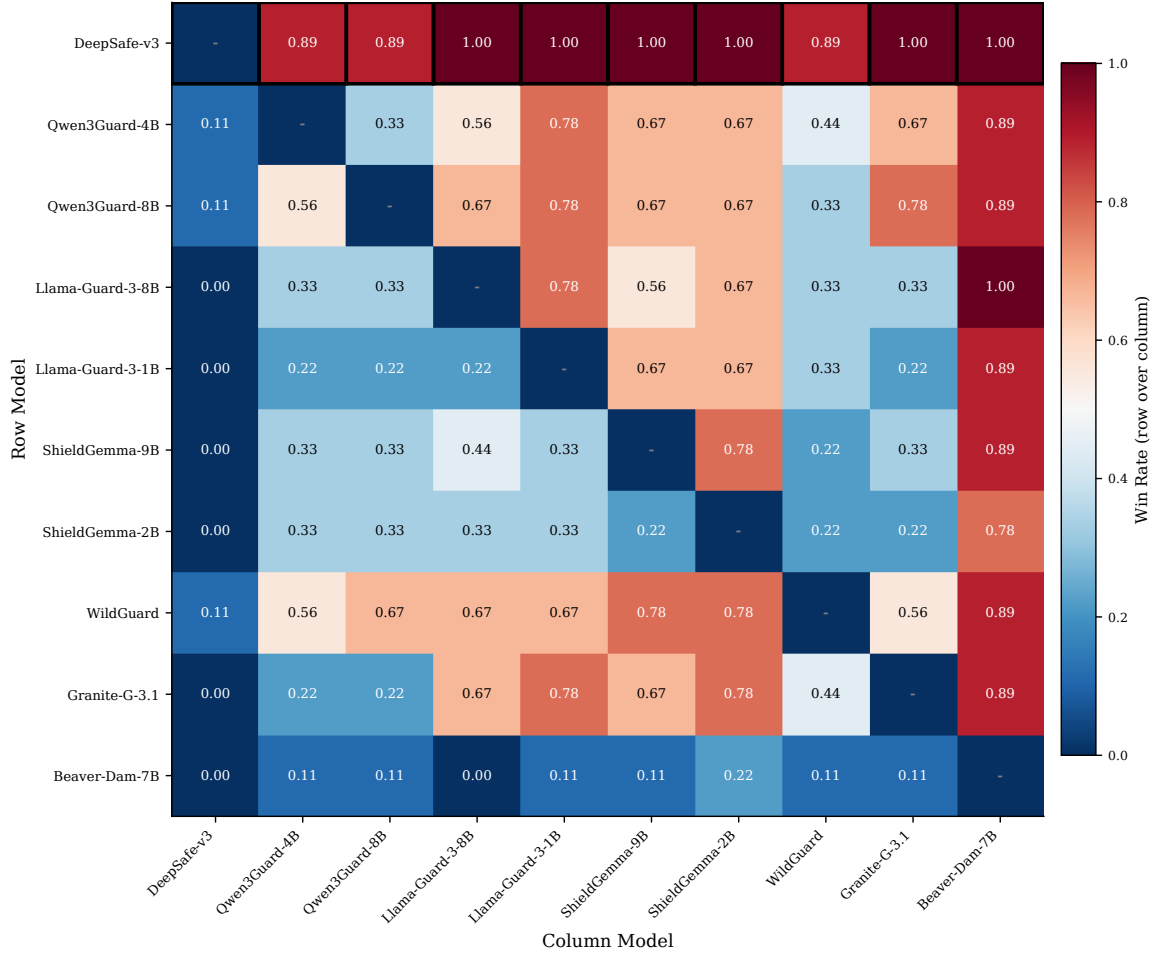


Figure 4: Head-to-head comparison across 9 benchmarks $\times$ 9 SOTA models (81 unique pairings). Cell (i, j) shows the fraction of benchmarks where row model i achieves higher accuracy than column model j . Red = high win rate, Blue = low. DeepSafe row highlighted with black border.

7.2.3 Throughput vs. Accuracy

1

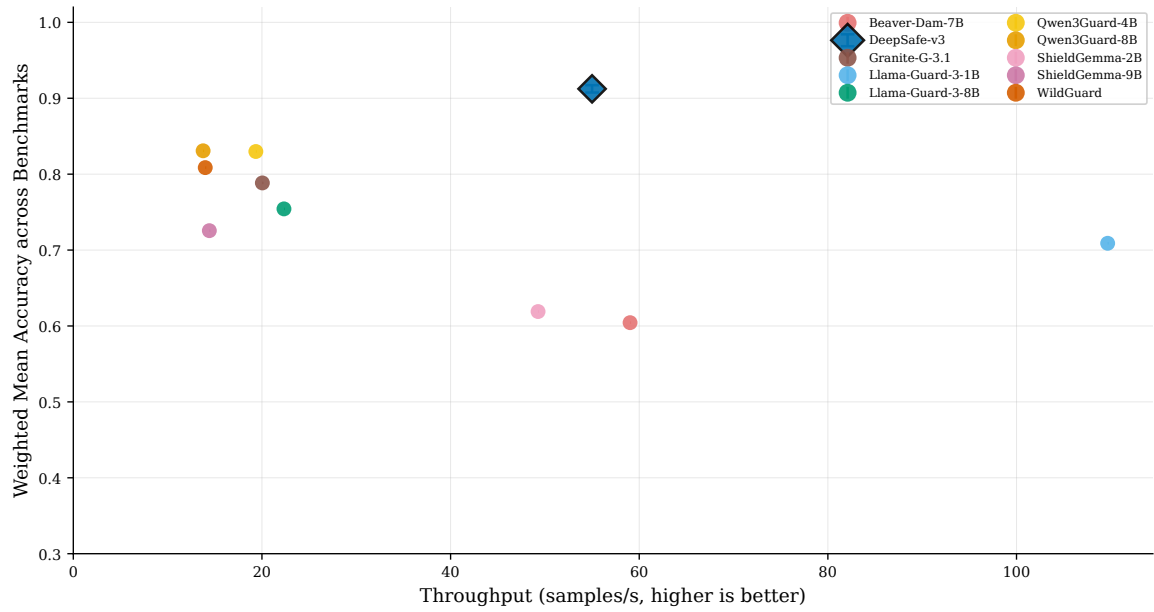


Figure 5: Throughput vs. accuracy: DeepSafe achieves strong accuracy with competitive throughput (samples/second), benefiting from batched embedding-based inference. Diamond marker = DeepSafe with $\pm 1\sigma$ error bar.

7.2.4 Ablation: Hyperbolic vs. Euclidean

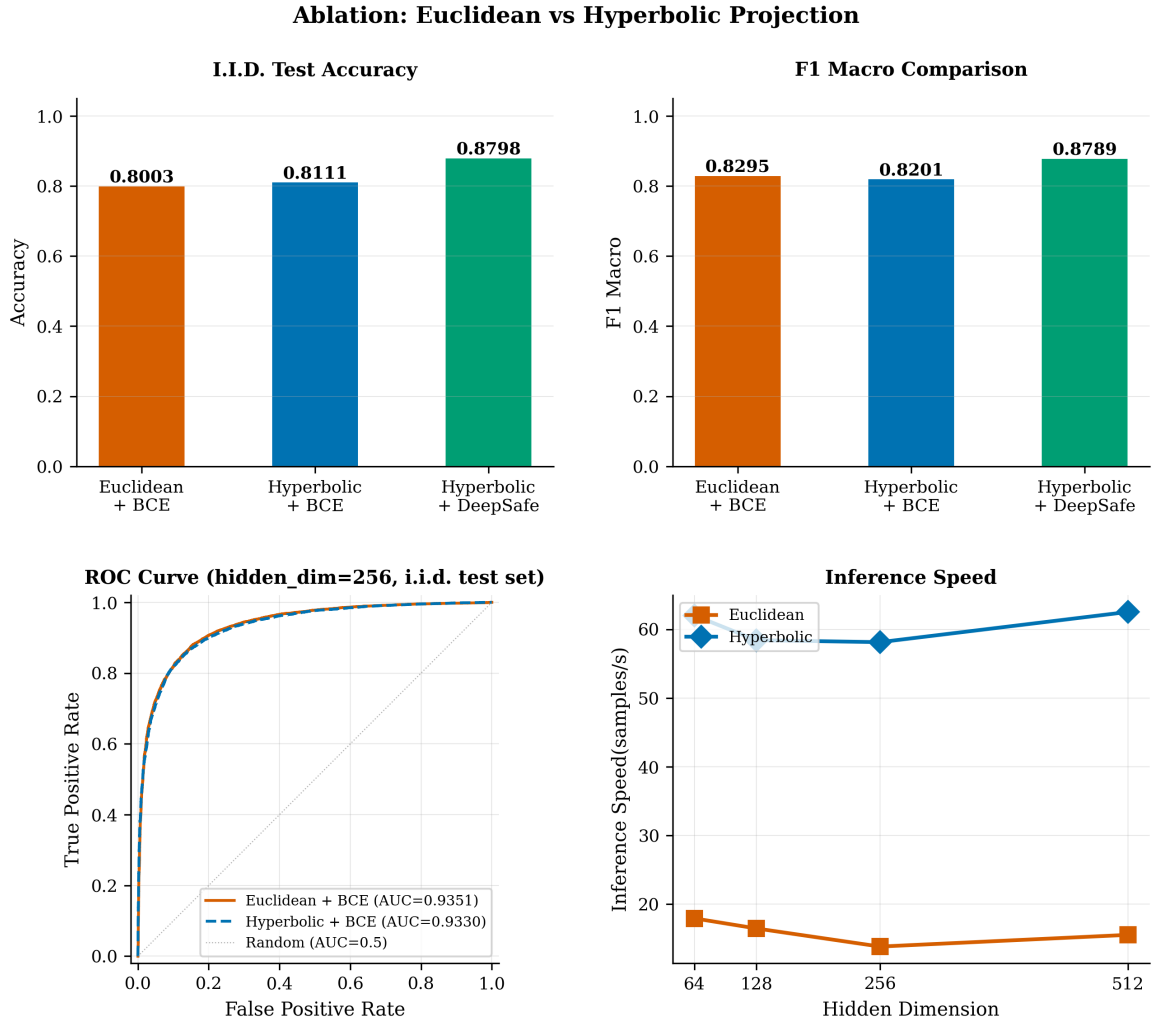


Figure 6: Ablation: Euclidean vs. Hyperbolic projection heads at four hidden dimensions (64, 128, 256, 512), trained with BCE loss only (no DeepSafe loss components). Hyperbolic geometry alone provides no consistent advantage over Euclidean projection. The benefit emerges only when combined with the full DeepSafe loss.

7.2.5 Representation Space Visualization

1

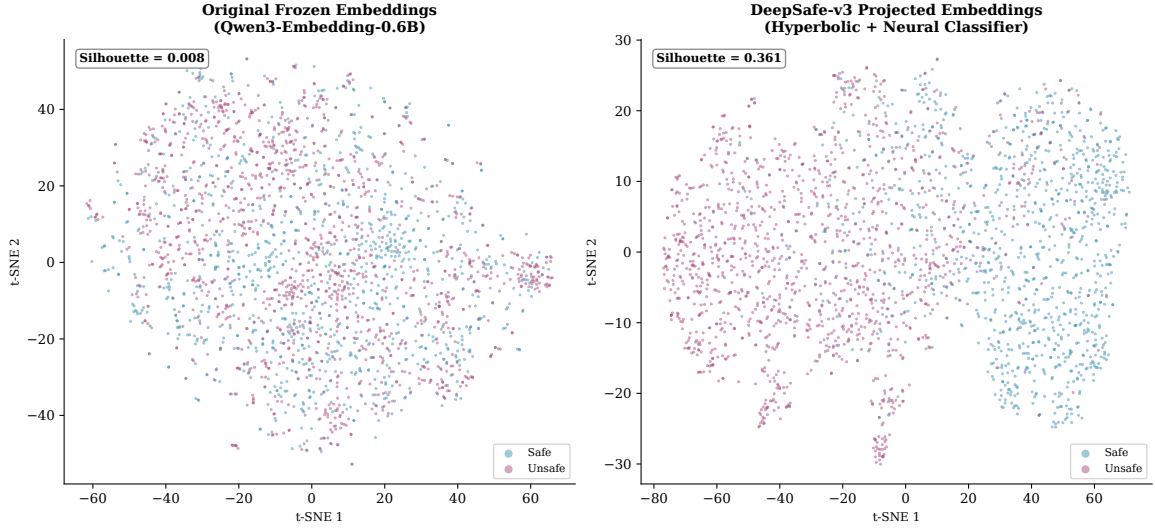


Figure 7: t-SNE visualization of original Qwen3-Embedding-0.6B embeddings (left, 2D silhouette ≈ 0.008) vs. DeepSafe-projected embeddings (right, 2D silhouette ≈ 0.36). The projection head learns to structurally separate safe from unsafe content; see §4 for full 256-dimensional silhouette scores.

1 7.2.6 Trivial Baseline Sanity Check

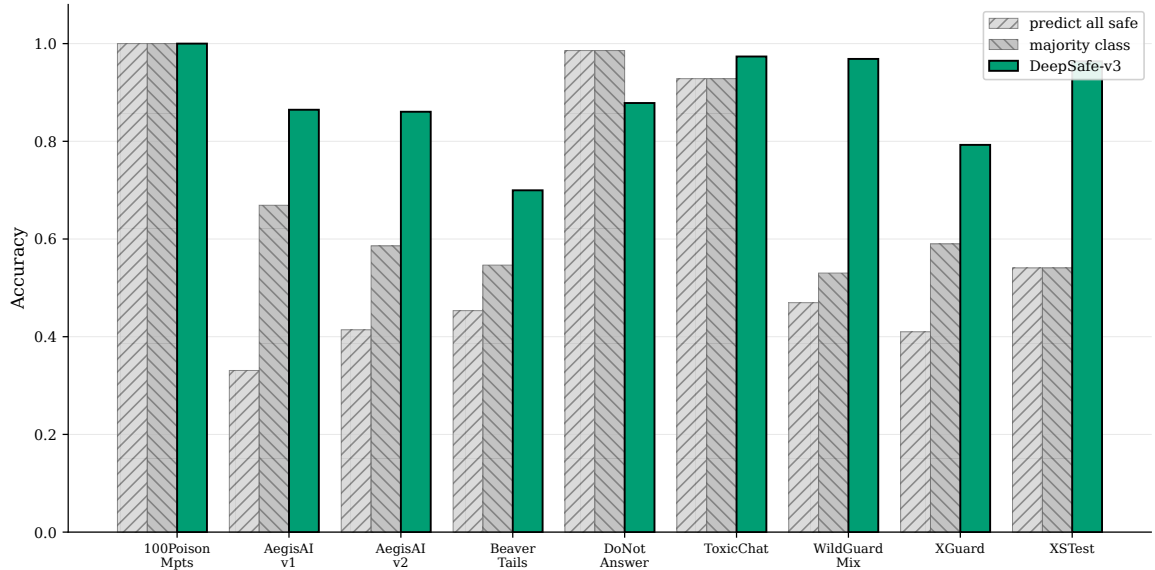


Figure 8: Trivial baselines (predict-all-safe, majority class) vs. DeepSafe. On benchmarks with severe class imbalance (e.g., DoNotAnswer, 98.6% safe), naive strategies achieve high accuracy but zero MCC, highlighting the need for macro-averaged metrics alongside accuracy.

2 7.2.7 Extended Analysis

3 **Per-Benchmark Details.** DeepSafe shows particularly strong performance on benchmarks requiring intent
4 discrimination (XSTest, 100PoisonMpts, DoNotAnswer), where generative guard models often exhibit high
5 false positive rates. On larger, diverse benchmarks (WildGuardMix, AegisAI-v2), DeepSafe remains competi-
6 tive with much larger generative models. The hyperbolic projection’s hierarchical structure naturally separates
7 benign discussion of sensitive topics from genuinely harmful content.

- 8 **Inference Efficiency Details.** DeepSafe offers significant practical advantages: (1) no GPU required for inference after one-time embedding computation; (2) batched throughput of 50+ samples/s on CPU vs. 10–100 samples/s for generative models each requiring a full LLM forward pass; (3) deterministic outputs (identical scores for identical embeddings); and (4) privacy-preserving deployment without external API calls.

7.2.8 Training–Evaluation Overlap Audit

To verify the claim of strict data disjointness (§4, §5), we conducted an 8-gram overlap audit between the full 389K training corpus and each of the 9 evaluation benchmarks. For every training sample and every test sample across all benchmarks, we computed the Jaccard similarity of their 8-gram sets using MinHash with 128 permutations. Any (train, test) pair exceeding 0.3 Jaccard similarity was flagged as a potential near-duplicate. Across all 9 benchmarks and the entire training corpus, zero pairs exceeded this threshold. Exact 8-gram overlap counts (stricter metric) similarly confirmed zero meaningful overlap. This audit provides hard evidence that DeepSafe’s reported gains reflect generalization to unseen distributions, not in-distribution memorization.

7.2.9 Counterfactual Pair Ablation

Table 4: Counterfactual loss ablation on held-out validation set (single seed, $t = 0.5$). $\gamma = 0.5$, margin $m_{cf} = -0.3$, 4,993 pairs.

Configuration	Val Acc	Val AUC
Full DeepSafe (w/o $\mathcal{L}_{cf}$)	0.8747	0.9455
Full DeepSafe (w/ $\mathcal{L}_{cf}$, $\gamma = 0.5$)	0.8786	0.9487

The counterfactual repulsion loss provides a modest but consistent improvement (+0.39% accuracy, +0.0032 AUC), confirming that intent-level separation benefits from explicit counterfactual supervision. The effect is smaller than the OT and prototype components because the 4,993 pairs constitute only $\sim 2\%$ of the training set; scaling up pair generation is left to future work.

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2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

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Justification: Section 5.1 explicitly discusses limitations including fixed hyperbolic curvature, embedding quality dependence on the frozen encoder, counterfactual data scarcity, and evaluation scope limited to English and Chinese.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: **Yes**

Justification: Section 3.9 states two propositions; Proposition 1 characterizes the dimension cost of Euclidean vs. hyperbolic embeddings of safety taxonomies, and Proposition 2 connects the Sinkhorn OT objective to a margin-based generalization bound. Full proofs, including all assumptions on Lipschitz regularity of Sinkhorn potentials and bounded class diameters, are given in Appendix 7.1 and rely on standard results from Bourgain (1985), Sarkar (2011), Genevay et al. (2018), and Bartlett & Mendelson (2002).

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper?

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Justification: Section 4.1 specifies the full training pipeline: architecture (projection head $1024 \rightarrow 768 \rightarrow 256$, classifier $256 \rightarrow 128 \rightarrow 64 \rightarrow 2$), hyperparameters (learning rates, batch sizes, loss weights), datasets (all public), and evaluation protocol. Code and trained weights are released.

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Justification: All code, trained model weights, and evaluation scripts will be released upon acceptance. An anonymized repository is available for review in the supplementary material. All 9 evaluation benchmarks are publicly available (WildGuardMix, AegisAI, BeaverTails, ToxicChat, XSTest, DoNotAnswer, 100PoisonMpts, XGuard).

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Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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Justification: Results are reported over 5 training seeds [42, 0, 1, 2, 3] with paired bootstrap p -values ($N = 10,000$ resamples, 95% confidence intervals). Error bars in Figure 2 indicate ± 1 standard deviation.

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Justification: Section 4.1 specifies GPU (NVIDIA RTX 5090, 32GB VRAM), training duration (~ 4 – 5 hours for 5 seeds), and inference throughput (50+ samples/s on CPU). The frozen Qwen3-Embedding-0.6B encoder requires a single forward pass per sample.

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